# Dataset Processing and Selection Rationale

BOMNI, PIROPO, and CEPDOF Datasets for Fisheye Pedestrian Detection

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## 1 Introduction

This document describes the unified processing pipeline applied to BOMNI [1], PIROPO [2], and CEPDOF [3] datasets for evaluating projection configurations in fisheye pedestrian detection. BOMNI and PIROPO use rotated bounding box annotations provided by Tamura et al. [4]; CEPDOF uses annotations from its original authors in COCO-like JSON format. Despite the different source formats, all three datasets are converted to the same standard JSON representation and follow the same two-step processing workflow with manual quality verification to ensure evaluation integrity.

### 1.1 Acronyms and Abbreviations

- **AP**: Average Precision
- **BOMNI**: Bogazici Omni-directional and Multi-Camera Network Images
- **CEPDOF**: Challenging Events for Person Detection from Overhead Fisheye images
- **COCO**: Common Objects in Context (object detection benchmark dataset)
- **GT**: Ground Truth
- **IoU**: Intersection over Union
- **PIROPO**: People in Indoor ROoms with Perspective and Omnidirectional cameras
- **TBD**: To Be Determined (values pending manual verification completion)
- **VOC**: Visual Object Classes (Pascal VOC annotation format)
- **XML**: eXtensible Markup Language
- **JSON**: JavaScript Object Notation

## 2 Common Processing Pipeline

Both BOMNI and PIROPO undergo identical processing steps, ensuring methodological consistency for fair cross-dataset comparison.

### 2.1 Step 1: Automated Preparation

#### 2.1.1 Annotation Format Conversion

**Input**: Tamura XML annotations in repurposed Pascal VOC format

**Process**:

1. Parse XML files to extract rotated bounding box parameters
2. Decode center and dimensions from xmin/ymin/xmax/ymax fields:
   - $center_x = (xmin + xmax)/2$
   - $center_y = (ymin + ymax)/2$
   - $width = xmax - xmin$
   - $height = ymax - ymin$
3. Compute rotation angle: $\theta = \arctan 2(dx, -dy)$ where $(dx, dy)$ is the vector from fisheye center to bbox center
4. Convert to standard JSON format with explicit fields: `center_x`, `center_y`, `width`, `height`, `angle`, `class_name`

**Output**: JSON annotation files with precomputed rotation angles

#### 2.1.2 Why Tamura's Format Requires Conversion

**Tamura's format limitation**: The original Pascal VOC XML format does *not* include an explicit rotation angle field. Tamura et al. repurposed the standard Pascal VOC bounding box fields (xmin, ymin, xmax, ymax) to encode the center position and dimensions of rotated bounding boxes, but the rotation angle itself is *omitted*.

**Implicit angle encoding**: Tamura assumes all pedestrian bounding boxes follow the fisheye radial distortion pattern. The rotation angle can be reconstructed geometrically:

$$\theta = \text{arctan2}(dx, -dy) \quad \text{where} \quad \begin{cases} dx = \text{bbox\_center}_x - \text{fisheye\_center}_x \\ dy = \text{bbox\_center}_y - \text{fisheye\_center}_y \end{cases} \tag{1}$$

This $(dx, dy)$ vector points from the fisheye center to the pedestrian's center, and its angle determines the bbox rotation (height axis aligned radially).

**Why this design is problematic**:

1. **Computational redundancy**: The rotation angle must be recalculated via `arctan2()` *every single time* annotations are loaded—during training (thousands of epochs $\times$ thousands of images), evaluation, and visualization. This represents massive redundant computation for a value that never changes.
2. **Format ambiguity**: Standard Pascal VOC parsers will misinterpret the annotations as axis-aligned boxes, yielding incorrect results unless the user knows about Tamura's repurposing convention.
3. **Lack of flexibility**: The implicit angle encoding enforces the radial constraint. If non-radial rotations are needed (e.g., for data augmentation or correcting annotation errors), the format cannot represent them.

**Why Tamura chose this format**: Likely reasons include:

- **Pascal VOC compatibility**: Adhering to an established annotation standard (Pascal VOC) for backwards compatibility with existing tools, despite the standard not supporting rotation.
- **Legacy tooling**: Reusing existing annotation tools that output Pascal VOC format rather than developing custom tooling.
- **Enforcing radial constraint**: Ensuring all annotators follow the fisheye radial pattern by making angle calculation automatic rather than manual.

**Our conversion rationale**: We address these limitations by:

1. **Precomputing angles once**: Calculate $\theta = \arctan 2(dx, -dy)$ during the one-time conversion step and store it explicitly in JSON. This eliminates redundant trigonometric operations during training/evaluation (compute once, use thousands of times).
2. **Explicit, self-documenting format**: Our JSON schema (`center_x`, `center_y`, `width`, `height`, `angle`, `class_name`) is immediately clear and requires no special knowledge to interpret.
3. **Unified cross-dataset format**: BOMNI and PIROPO annotations become identical in structure, enabling shared data loading code and consistent evaluation pipelines.
4. **Performance optimization**: Faster data loading during training (critical for large-scale experiments with multiple projection configurations).

**Computational savings**: For a dataset with $N$ images evaluated over $E$ epochs during training, our conversion eliminates $N \times E$ `arctan2()` calls. For PIROPO with 3,089 images evaluated 100 times (training and hyperparameter tuning), this saves approximately 309,000 trigonometric operations.

### 2.1.3 Frame Extraction and Filtering

**Challenge**: Original datasets contain thousands of frames per sequence, but Tamura only annotated sparse keyframes ($\sim$10% of total frames).

**Process**:

1. Extract or organize frames from videos/downloads
2. Identify annotated frames by matching filenames with Tamura annotation files
3. **Delete all unannotated frames** (keeps only frames with verified ground truth)
4. Organize frames to match annotation directory hierarchy

**Rationale**: Unannotated frames may contain unlabeled pedestrians (false ground truth negatives) or be truly empty. Including them without verified annotations would compromise evaluation integrity. We conservatively use only frames with explicit ground truth labels.

#### 2.1.4 Visualization Generation

**Process**:

1. Load each annotated frame and corresponding JSON annotations
2. Draw rotated bounding boxes on images using OpenCV
3. Save visualizations to dedicated output directory

**Output**: Visual representations of all annotations for human inspection

### 2.2 Manual Quality Review

**User task**:

1. Inspect all visualization images
2. Identify annotation errors:
   - Incorrect bbox position (misaligned with pedestrian)
   - Wrong size (too large/small, loose vs. tight fit inconsistency)
   - Incorrect rotation angle (not aligned with radial direction)
   - False positives (background objects mislabeled)
   - Missing detections (pedestrian visible but not annotated)
3. **Delete visualization images with incorrect annotations**
4. Keep only visualizations representing correct ground truth

**Critical requirement**: User deletes *only visualization images*, never annotation files directly. The automated cleanup step (Step 2) handles annotation removal.

### 2.3 Step 2: Automated Cleanup

**Process**:

1. Scan remaining visualization images (after user deletion)
2. Identify annotation files lacking corresponding visualizations (user deleted them)
3. **Delete orphaned annotation JSON files and frame images**
4. Regenerate final clean visualizations from verified annotations

**Output**: Production-ready dataset containing only manually verified correct annotations

**Rationale**: Manual verification filters annotation noise. Tamura annotations, like all manual labeling efforts, contain errors. Our BOMNI verification revealed a 25.5% error rate. Evaluating on uncorrected annotations yields misleading metrics influenced by annotation noise rather than true model performance.

# 3 BOMNI Dataset Processing

## 3.1 Dataset Overview

- **Source**: Bogazici Omni-directional and Multi-Camera Network Images
- **Environment**: Single indoor room, ceiling-mounted fisheye camera
- **Resolution**: 640×480 pixels
- **Fisheye center**: (320, 240)
- **Sequences**: scenario1 with 4 videos (top-0, top-1, top-2, top-3)
- **Tamura annotations**: 337 frames, 1,122 person instances (initial)

## 3.2 BOMNI-Specific Steps

### 3.2.1 Frame Extraction from Videos

- Input: 4 MP4 video files (top-0.mp4 through top-3.mp4)
- Extract all frames using OpenCV (cv2.VideoCapture)
- Naming: 4-digit serial numbers (0001.jpg, 0002.jpg, ...)
- Output: Thousands of frames per sequence

### 3.2.2 Cleanup and Conversion

- Delete unannotated frames (reduces from thousands to 337 annotated frames)
- Convert 337 Tamura XML files to standard JSON format
- Generate visualizations for all 337 frames

## 3.3 Manual Verification Results

- **Frames reviewed**: 337
- **Incorrect annotations identified**: 86
- **Error rate**: 25.5%
- **Error types**: Wrong position, inaccurate size, incorrect orientation, false positives
- **Final verified dataset**: 245 frames, 834 person instances (BOMNI-production)

### 3.4 Sequence Aggregation

**Strategy**: Combine all 4 sequences (top-0 through top-3) into unified BOMNI dataset
**Justification**:

- Per-sequence sample size insufficient (∼60 frames each) for reliable AP/precision/recall
- Identical camera position across sequences (same fisheye center, distortion model)
- Aggregation captures pedestrian diversity (appearance, pose, scale variation)
- Research goal: evaluate projection generalization, not sequence-specific tuning

### 3.5 BOMNI Limitations

1. **Limited scene variability**: Single room, constant lighting, uniform background
2. **Low resolution**: 640×480 (less detail than PIROPO's 800×600)
3. **Repetitive frames**: Near-duplicates from static pedestrians
4. **Small dataset**: 245 images (modest compared to COCO's 118k training images)
5. **Annotation noise**: 1–2% residual error expected even after manual correction

## 4 PIROPO Dataset Processing

### 4.1 Dataset Overview

- **Source**: People in Indoor ROoms with Perspective and Omnidirectional cameras
- **Rooms**: Room A (15×10m), Room B (8×5m)
- **Cameras**: 8 in Room A (3 omni + 5 perspective), 2 in Room B (1 omni + 1 perspective)
- **Resolution**: 800×600 pixels
- **Fisheye center**: (400, 300)
- **Tamura annotations**: 3,089 frames, 3,550 person instances (omni cameras only)

### 4.2 PIROPO-Specific Challenges

#### 4.2.1 Massive Frame Archives

- Each camera folder: 2.8–4.9 GB (thousands of frames per sequence)
- Tamura annotated only sparse keyframes (∼10% of available frames)
- Unannotated frames drastically outnumber annotated ones

**Solution**: Extract *only* frames matching Tamura annotation timestamps (identified by ISO 8601 filenames, e.g., `2015-05-12T12-17-03.626Z.jpg`). Reduces dataset from ∼15 GB to ∼1 GB.

#### 4.2.2 Hierarchical Folder Structure

- Original PIROPO: `Room A/omni_3A/omni_3A/omni3A_training/`
- Tamura annotations: `Room_A/omni_3A/omni_3A_training/`
- Differences: Spaces vs. underscores, duplicate camera folders, sequence naming

**Solution**: Reorganize extracted frames to match Tamura's flattened structure with standardized naming (`Room_A/{camera}/{sequence}/`).

### 4.3 Camera Selection

#### 4.3.1 Excluded: Perspective Cameras

**Cameras excluded**: conv_4A through conv_8A (Room A), conv_2B (Room B)

**Justification**: Our research objective is fisheye pedestrian detection. Perspective cameras exhibit minimal radial distortion and do not require gnomonic projection transformations. Including perspective data would dilute evaluation metrics with out-of-scope samples.

#### 4.3.2 Included: Omnidirectional Cameras Only

**Cameras annotated by Tamura**:

- Room_A: omni_1A, omni_2A, omni_3A
- Room_B: omni_1B

**Total**: 4 fisheye cameras, 3,089 frames (pre-verification)

### 4.4 Sequence Selection

#### 4.4.1 Excluded: Unannotated Sequences

**Sequences excluded**:

- test4 through test12 (marked "SEQ" in PIROPO docs = sequences exist but no ground truth)
- training_seats (sitting pedestrians, not annotated by Tamura)

**Justification**: Quantitative evaluation requires ground truth. Sequences without Tamura annotations cannot contribute to precision, recall, AP, or IoU-based metrics.

#### 4.4.2 Excluded: Empty Background Sequences

**Sequences excluded**: training_illum, training_bg, training_additional_bg

**Content**: Only background with illumination/appearance changes, no pedestrians

**Justification**: These sequences were provided for background modeling in classical motion-based detection (e.g., background subtraction). Our YOLO-based detector does not require background modeling. Frames without pedestrians contribute zero true positives and zero false negatives, providing no evaluation signal.

#### 4.4.3 Included: Annotated Pedestrian Sequences

**Sequences annotated by Tamura**:

- **training**: Person walking exhaustively around room (3 people)
- **test2**: Up to 3 people simultaneously walking
- **test3**: 1 new person (not in training) walking

### 4.5 Camera Aggregation

**Multi-camera structure**:

- 4 physical cameras at different ceiling positions
- Same sequences recorded *simultaneously* from multiple viewpoints
- Same pedestrians captured from different angles and distances

**Strategy**: Combine all 3,089 annotations from all 4 cameras into unified PIROPO dataset
**Justification**:

1. **Statistical robustness**: Larger sample size (3,089 vs. ∼750 per camera) for reliable metric estimates
2. **Viewpoint diversity**: Multiple angles stress-test projection configurations across varying radial distortion patterns
3. **Research alignment**: Goal is optimal configuration for *general* fisheye detection, not camera-specific tuning
4. **Consistency with BOMNI**: BOMNI aggregates sequences; PIROPO aggregates cameras. Both maximize data for robust evaluation.

**Secondary analysis** (optional): Per-camera performance breakdown as sanity check for camera-specific issues.

### 4.6 Manual Verification (Pending)

Following the same protocol as BOMNI:

1. Visualize all 3,089 frames
2. Manual review and deletion of incorrect annotations
3. Automated cleanup
4. Expected error rate: 10–25% (based on BOMNI experience)

## 5 CEPDOF Dataset Processing

### 5.1 Dataset Overview

- **Source**: Challenging Events for Person Detection from Overhead Fisheye images [3]
- **Environment**: Single indoor classroom, overhead-mounted fisheye camera
- **Resolution**: 2,048×2,048 pixels (Lunch1–3, Edge_cases) and 1,080×1,080 pixels (High_activity, All_off, IRfilter, IRill)
- **Fisheye center**: Image center (auto-computed as $width/2$, $height/2$)
- **Sequences**: 8 videos covering diverse challenging scenarios (crowded scenes, body occlusions, unusual poses, low-light conditions)
- **Scale**: 25,504 annotated frames, 173,074 person instances across all sequences

### 5.2 Annotation Format

CEPDOF annotations follow a COCO-like JSON format, with one JSON file per video sequence. Each bounding box is represented as:

$$\text{bbox} = [c_x,\ c_y,\ w,\ h,\ d] \tag{2}$$

where $c_x, c_y$ are center coordinates, $w, h$ are width and height (constrained to $w \leq h$), and $d$ is the **clockwise rotation angle from the vertical axis** (pointing upward), in degrees, with $d \in [-90, +90]$.

This is a direct, explicit representation — no trigonometric derivation is needed. Conversion to our standard per-frame JSON format is a straightforward field mapping:

| CEPDOF field | Standard format field |
|---|---|
| $c_x$ | `center_x` |
| $c_y$ | `center_y` |
| $w$ | `width` |
| $h$ | `height` |
| $d$ | `angle` |
| (always person) | `class_name` |

The $w \leq h$ constraint and angle clamping are normalized once during conversion so the pipeline never needs to handle them.

### 5.3 Distinction from BOMNI and PIROPO: Non-Radial Bounding Boxes

**Critical difference**: Unlike BOMNI and PIROPO, where Tamura enforces a *radial alignment* constraint (the major axis of every bounding box points toward the fisheye center), CEPDOF annotations place **no such constraint**. Bounding boxes represent the actual body orientation of each person as observed from overhead, which may or may not be radially aligned.

In BOMNI and PIROPO (Tamura format), the angle is entirely determined by the spatial position of the pedestrian relative to the fisheye center:

$$\theta = \text{arctan2}(dx, -dy), \quad dx = c_x - f_x,\ dy = c_y - f_y \tag{3}$$

This enforces that every annotated pedestrian's body axis points radially away from the camera. While this assumption holds approximately for standing pedestrians, it breaks for people walking tangentially, sitting, or in unusual poses.

CEPDOF, by contrast, stores the independently observed angle $d$, which reflects the person's true body orientation regardless of their position in the image.

> **Note on orientation convention and IoU**: The $w \leq h$ constraint introduces a 90° ambiguity in angle representation (swapping $w$ and $h$ and rotating by 90° produces the same physical rectangle). For IoU-based evaluation (mAP, precision, recall), this ambiguity is irrelevant: intersection and union surfaces depend only on the physical rectangle, not on its parametrization. As long as the bounding box correctly represents the physical region occupied by the pedestrian, the IoU computation is unaffected by which (w, h, $d$) parametrization is used. This ambiguity would only matter if the angle $d$ is directly evaluated as a target metric, which is not the case in our evaluation protocol.

### 5.4 Implications for Backprojection

The non-radial alignment of CEPDOF annotations has an important methodological consequence for our backprojection pipeline. Our current backprojection method forces the height axis of every backprojected bounding box to point radially toward the fisheye center — a design choice inherited from the Tamura annotation convention of BOMNI and PIROPO.

CEPDOF provides a natural testbed to investigate whether this radial alignment is beneficial or harmful:

- **Radially-aligned backprojection** (current): Assumes every detected pedestrian stands upright relative to the fisheye center. Simple and consistent with Tamura-annotated datasets.
- **Free-orientation backprojection**: Preserves the detected orientation from the composite (perspective) image, transformed geometrically without forcing radial alignment.

Since CEPDOF ground truth does not enforce radial alignment, comparing these two backprojection strategies on CEPDOF will reveal whether the radial alignment assumption helps, hurts, or makes no significant difference in practice. This constitutes an additional experimental axis beyond projection configuration search.

### 5.5 Frame Filtering

CEPDOF provides annotations for nearly all frames: 25,358 out of 25,504 total frames are annotated (146 unannotated frames, typically empty scenes). Our preparation pipeline copies only annotated frames to the production folder, discarding the 146 unannotated ones.

### 5.6 Manual Verification

CEPDOF annotations are produced by the original dataset authors (Boston University VIP Laboratory) using spatiotemporal tracking, which generally yields higher consistency than the third-party Tamura annotations used for BOMNI and PIROPO. Nevertheless, we apply the same manual verification protocol for methodological consistency:

1. Sample $N$ frames per sequence (evenly spread across the full sequence to ensure representative coverage)

2. Visually inspect rotated bounding box overlays
3. Delete visualization images for frames with incorrect annotations
4. Run automated cleanup to remove corresponding annotation and frame files

**Sampling strategy**: Given the large scale of CEPDOF (25,358 frames), reviewing all frames is impractical. We therefore sample a configurable number of frames per sequence (default: 100) using uniform stride $s = \lfloor N_{\text{total}}/N_{\text{sample}} \rfloor$, ensuring that selected frames span the entire sequence rather than clustering at the beginning.

# 6 Cross-Dataset Comparison

| **Characteristic** | **BOMNI** | **PIROPO** | **CEPDOF** |
|---|---|---|---|
| Cameras (fisheye) | 1 | 4 | 1 |
| Sequences | 4 | 12 | 8 |
| Resolution | 640×480 | 800×600 | 2048×2048 / 1080×1080 |
| Fisheye center | (320, 240) | (400, 300) | auto (image center) |
| Annotated frames | 337 | 3,089 | 25,358 |
| Person instances | 1,122 | 3,550 | 173,074 |
| Verified frames | 245 | 3,004 | TBD |
| Annotation error rate | 25.5% | 2.75% | TBD |
| Annotation source | Tamura et al. | Tamura et al. | Original authors |
| Annotation format | Tamura XML | Tamura XML | COCO JSON |
| Radial alignment | Enforced | Enforced | Free (actual body orientation) |
| Environment | Single room | 2 rooms | Single classroom |
| Scene complexity | Uniform | Cluttered | Crowded, low-light, occlusion |
| **Processing pipeline** | **Identical (two-step with manual verification)** | | |

Table 1: Cross-dataset comparison. All three datasets follow the same two-step processing workflow with manual verification.

# 7 Methodological Consistency and Fair Evaluation

## 7.1 Why Consistency Matters

Applying identical processing to BOMNI and PIROPO ensures:

- Differences in projection configuration performance reflect *true dataset characteristics* (scene complexity, pedestrian scale, lighting) rather than processing artifacts
- Cross-dataset validation of optimal configurations (if Config A wins on both datasets, stronger evidence of generalization)
- Fair comparison: Same annotation quality standard (manual verification), same evaluation protocol

### 7.2 Conservative Data Inclusion

Our approach prioritizes **evaluation integrity over dataset size**:

- Exclude unannotated frames (avoid false negatives from missing labels)
- Exclude incorrect annotations (avoid rewarding wrong predictions)
- Exclude irrelevant data (perspective cameras, empty backgrounds)
- Result: Smaller but *cleaner* datasets with verified ground truth

### 7.3 Limitations and Mitigation

**Small dataset sizes**: BOMNI (245 images) and PIROPO (∼3,000 images) are modest compared to COCO (118k) or ImageNet (1.2M).

**Mitigation strategies**:

1. Focus on *relative* comparisons (projection configuration rankings) rather than absolute AP values
2. Cross-dataset validation (consistency across BOMNI and PIROPO increases confidence)
3. Conservative claims (acknowledge generalization limits beyond indoor fisheye scenarios)
4. Statistical significance testing (if sample size permits)

## 8 Summary

Both BOMNI and PIROPO datasets undergo rigorous two-step processing with manual quality verification:

1. **Step 1**: Automated extraction, filtering, conversion, and visualization
2. **Manual review**: Human inspection and deletion of incorrect annotations
3. **Step 2**: Automated cleanup and final dataset generation

Final production datasets contain only verified correct annotations:

- **BOMNI-production**: 245 frames, 834 person instances (25.5% error rate)
- **PIROPO-production**: 3,004 frames, 3,465 person instances (2.75% error rate)
- **CEPDOF-production**: TBD frames (pending manual verification), ∼173,000 person instances

This methodological rigor ensures evaluation metrics reflect true projection configuration performance rather than annotation noise or processing inconsistencies.

# References


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